

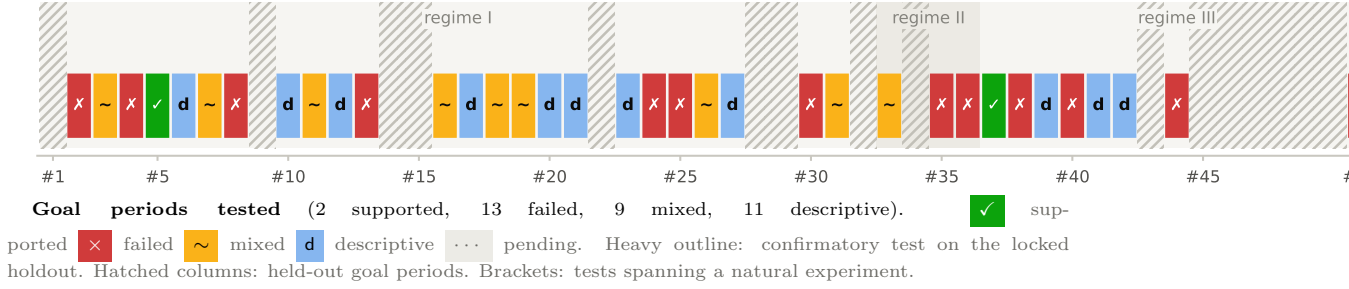
H03: Swarm activity is self-exciting, and its criticality is set by the goal's coupling mode

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04
(Dated:)

Complete 40% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. The swarm is clearly subcritical ($n \approx 0.2\text{--}0.4$) and its self-excitation does not depend on the goal's coupling mode; only a small, fast, reply-like cross-triggering survives its null.

Status: running. Exploratory round 1 is complete (2026-10-03, non-holdout data only). (level: hypothesis · pipeline checks 71% · updated 2026-10-04)



WHAT HAPPENED

Hawkes fits to agent talk events in 35 non-holdout goal periods (282 days), each period fitted on its own with a schedule baseline to guard against spurious criticality. Predicted: shared-objective weeks run near critical (median $\hat{n} \geq 0.7$) and above free weeks, and #51 drifts toward $n = 1$ as its roster grows. All failed. Median $\hat{n}$ is 0.41 (range 0–0.76), bracketed at 0.2–0.4 by the baseline choice. Shared-objective weeks gave 0.32 vs 0.53 for free weeks ($p = 0.75$), $\hat{n}$ falls with swarm size ($\rho = -0.53$), and #51 moves *away* from criticality as it grows ($\rho = -0.64$). A constant baseline would have “confirmed” criticality ($\hat{n} \approx 0.99$), and pooled $\hat{n}$ cannot be separated from 10–30-min rate modulation. The robust positive: fast cross-agent triggering, $n_{\times} \approx 0.07$ at 10–30 s lags, about $4\times$ an agent-shift null ($p = 2 \times 10^{-9}$). Humans and the nudger directly drive only about 1.5% of events. No holdout run yet.

PHYSICAL MODEL

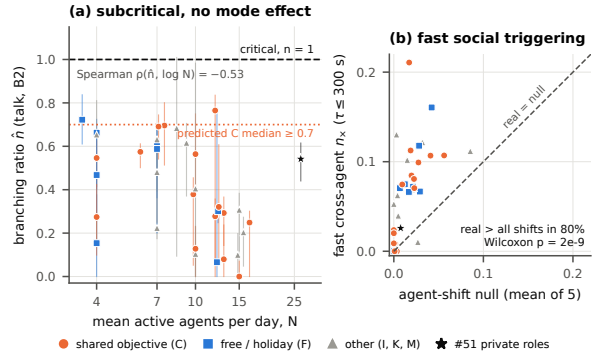
Model 09 (Hawkes self-exciting point process). Events: agent messages; one realization per village day. $\mu_B(t)$: schedule baseline (per-day level $\times$ 30-min shape, plus a kickoff bump); g_{exo} : response to human and nudger messages; ϕ : triggering kernel; n : branching ratio, the mean number of events each event directly triggers (critical at $n = 1$; $\langle s \rangle$: mean cascade size). Multivariate fit: λ_a , agent a 's intensity, with own and cross kernels; $\bar{m}$: active agents; $w_m^{\times}$: cross weight at timescale τ_m .

$$\lambda(t) = \mu_B(t) + \sum_{x < t} g_{\text{exo}}(t - x) + \sum_{t_j < t} \phi(t - t_j)$$

$$\phi(\tau) = n\beta e^{-\beta\tau}, \quad \langle s \rangle = 1/(1 - n)$$

$$\lambda_a(t) = c_a\mu(t) + \sum_{t_j \in a} \phi_{\text{self}}(t - t_j) + \sum_{t_j \notin a} \phi_{\times}(t - t_j)$$

$$n_{\times} = (\bar{m} - 1) \sum_{\tau_m \leq 300\text{s}} w_m^{\times}$$



Observables. (a) Branching ratio per goal period against swarm size, colored by coupling mode, with day-bootstrap 95% CIs. No period is near critical, shared-objective periods sit no higher than free ones, and $\hat{n}$ falls as the swarm grows. (b) Fast cross-agent excitation against an agent-shift null that breaks cross-agent timing: nearly every period sits above the diagonal. This is the one component that beats its strongest null.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A negative that matters a little: this swarm is nowhere near a runaway cascade, and neither more agents nor a shared goal pushes it there; bigger swarms trigger less per agent. Anyone worried about message storms gets mild

reassurance, but activity is set mostly by the scheduler and the roster, not by social contagion. The one robust number, each message drawing about 0.07 extra messages from others within half a minute, may just be agents waking when a message lands: a scaffold property. A

naive constant-baseline Hawkes fit would have reported near-criticality, an easy mistake to make. Practically: one pitfall and one small coupling constant.

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